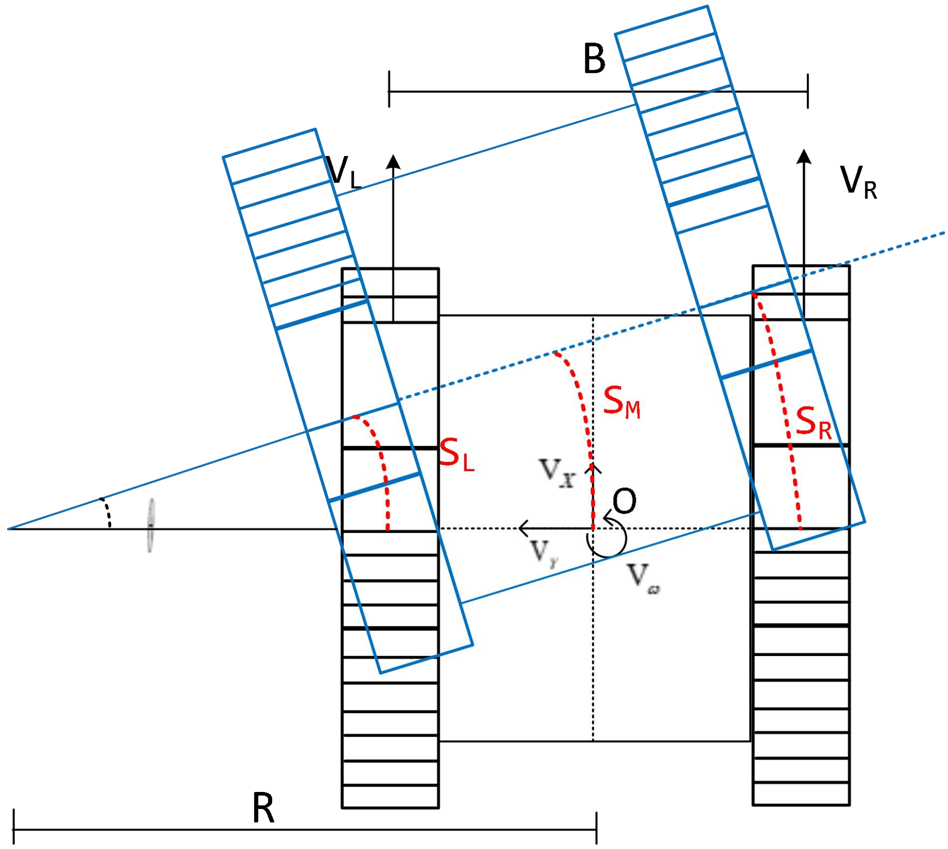
Track Chassis Kinematics

Track vehicle kinematics primarily focuses on examining the kinematic interactions between the chassis and individual wheel components of a track vehicle across various driving conditions. Once we comprehend the hardware structure and physical attributes mentioned earlier, let's delve into the kinematic analysis of the track vehicle. The figure below illustrates the kinematic model of the track vehicle, encompassing the following parameters:



1) **B:** On the track - wheelbase, the distance between the two driving wheels, typically measured in meters (m).

2) **R:** Turning radius (R) resulting from the robot moving forward and rotating simultaneously, typically measured in meters (m).

3) V_x: Forward and backward speed of the robot at point O. Forward speed is positive, usually measured in meters per second (m/s).

4) V_y: Target left and right speed of the robot at point O. Positive values indicate leftward movement, unit: m/s. Ignoring both rounds.

5) V_w: Target rotation speed of the robot around point O. Positive values indicate counterclockwise rotation, unit: rad/s.

6) V_L : Left wheel speed of the robot. Positive in the forward direction, typically measured in meters per second (m/s).

7) S_R : Speed of the robot's right wheel. Positive in the forward direction, typically measured in meters per second (m/s).

8) S_L : Path traveled by the left wheel within a certain time t.

9) S_M : Path traveled by the middle point O within a certain time t.

10) S_R : Path traveled by the right wheel within a certain time t.

11) θ : Angle of rotation of the robot within a certain time t, measured in radians.

Formula Calculation:

1.Their relationship is explored below, leading to the derivation of both the forward and inverse kinematics formulas for the track chassis (two-wheel differential type) robot. It is noted that the integral of speed over time equals the distance:

$$S_L = V_L * \mathrm{t} \quad , \quad S_M = V_x * \mathrm{t} \quad , \quad S_R = V_R * \mathrm{t}$$

2.The ratio of arc length to radius equals radians.

$$\theta = \frac{S_L}{R - \frac{B}{2}} = \frac{S_M}{R} = \frac{S_R}{R + \frac{B}{2}}$$

$$\theta = \frac{V_L * t}{R - \frac{B}{2}} = \frac{V_x * t}{R} = \frac{V_R * t}{R + \frac{B}{2}}$$

3. Simultaneously divide both sides of the formula by 't', i.e., integrate over time:

$$V_\omega = \frac{V_L}{R - \frac{B}{2}} = \frac{V_x}{R} = \frac{V_R}{R + \frac{B}{2}}$$

4. Solve the inverse kinematics equation based on the aforementioned formula. Given the robot's target speed V_x and V_ω, determine the drive wheel speed.

$$V_L = V_X - \frac{V_\omega * \mathrm{B}}{2}, V_R = V_X + \frac{V_\omega * \mathrm{B}}{2}$$

5. Utilize the inverse kinematics equation to solve the forward kinematics equation. Given the current speed of the drive wheel V_L and V_R, calculate the real-time speed of the robot V_x and V_ω

$$V_x = \frac{V_L + V_B}{2}, \quad V_\omega = \frac{-V_L + V_B}{B}$$